

AboutBits

Meet Up Talks

February 2024 | @ NOI Techpark Bruneck



GKN Powder Metallurgy – By the Numbers



PEOPLE

5,000+



MANUFACTURING LOCATIONS

27



CUSTOMERS

3,000+



SALES REVENUE

1 bn



 HOEGANAES



World #2 - 250,000 tons/year



 GKN SINTER METALS



World #1 - 10,000,000 pcs/day



 GKN ADDITIVE



Global Leader - 2M pcs/year

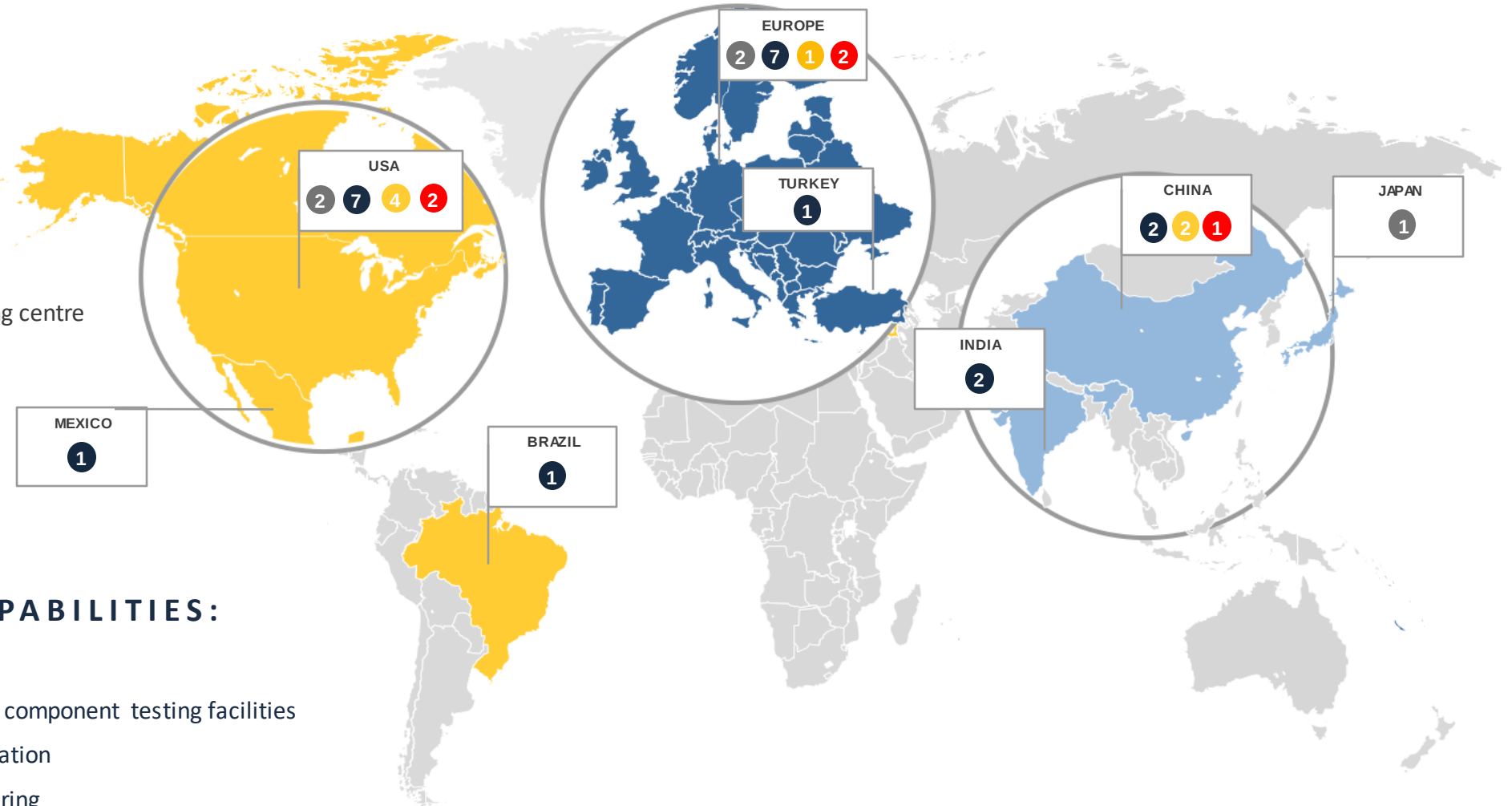
GKN Powder Metallurgy - Our Global Footprint

MANUFACTURING:

- Corporate sales office & engineering centre
- Sinter Metals
- Hoeganaes
- Additive

R&D CENTER CAPABILITIES:

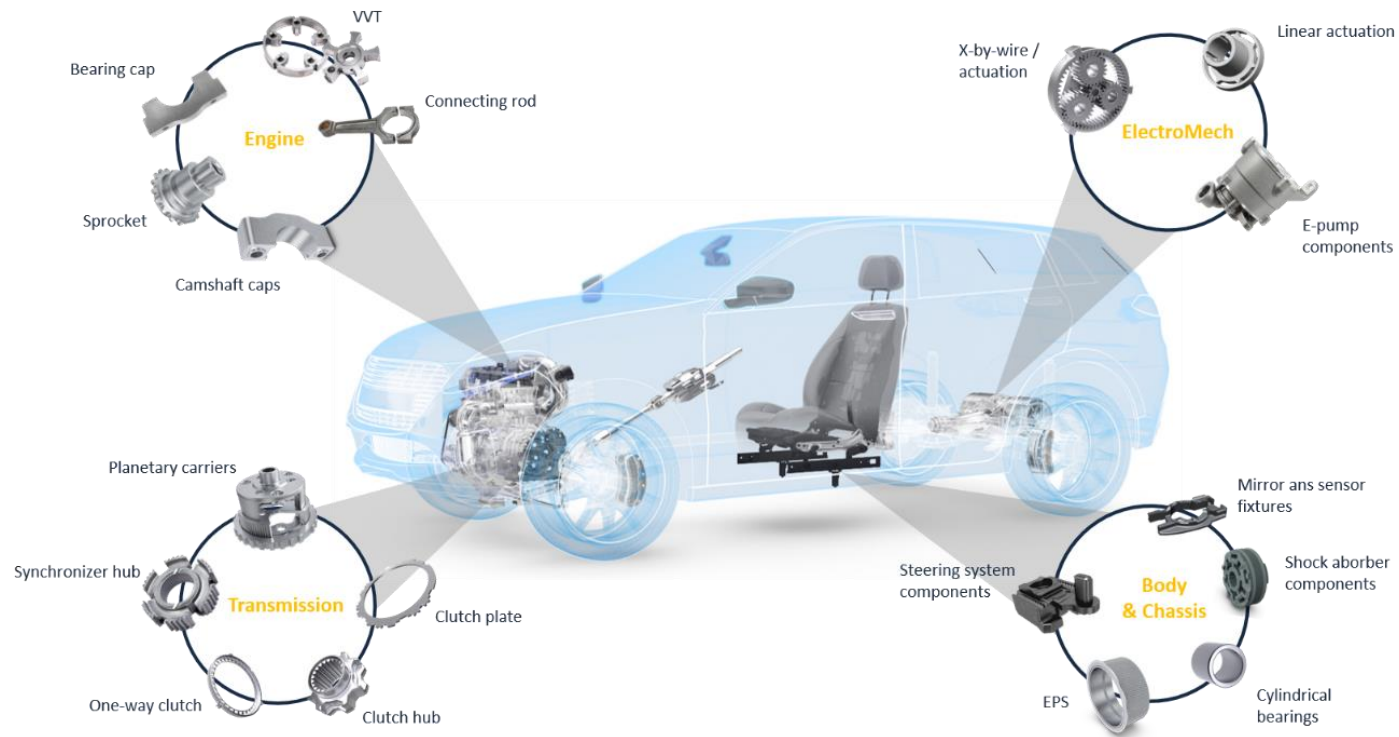
- > Powder development
- > Fully fledged material and component testing facilities
- > Topology & form optimization
- > Electromagnetic engineering



Key Powder Metal Applications & Components

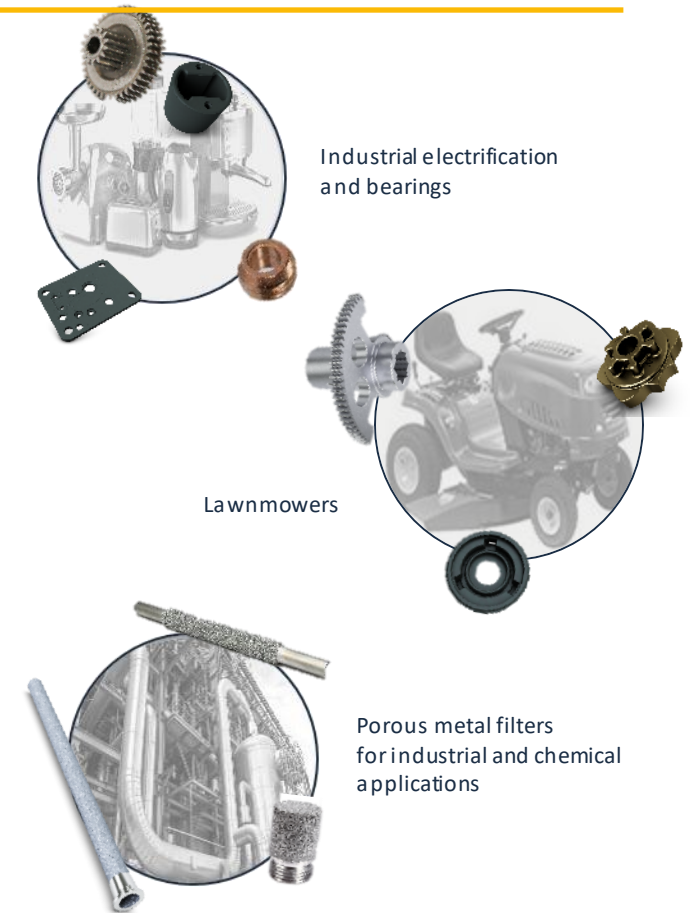
Key Automotive applications

(> 80% of Sinter Metals sales)



Key Industrial applications

(<20% of Sinter Metals sales)





GKN Powder Metallurgy
has created a digital
culture that drives value
and increases
competitive advantage

More than
1,800
connected
machines

6,500
digitally
connected
employees

Globally
deployed and
standardized
IT solutions

Over 800
digital
connected
measurement
stations

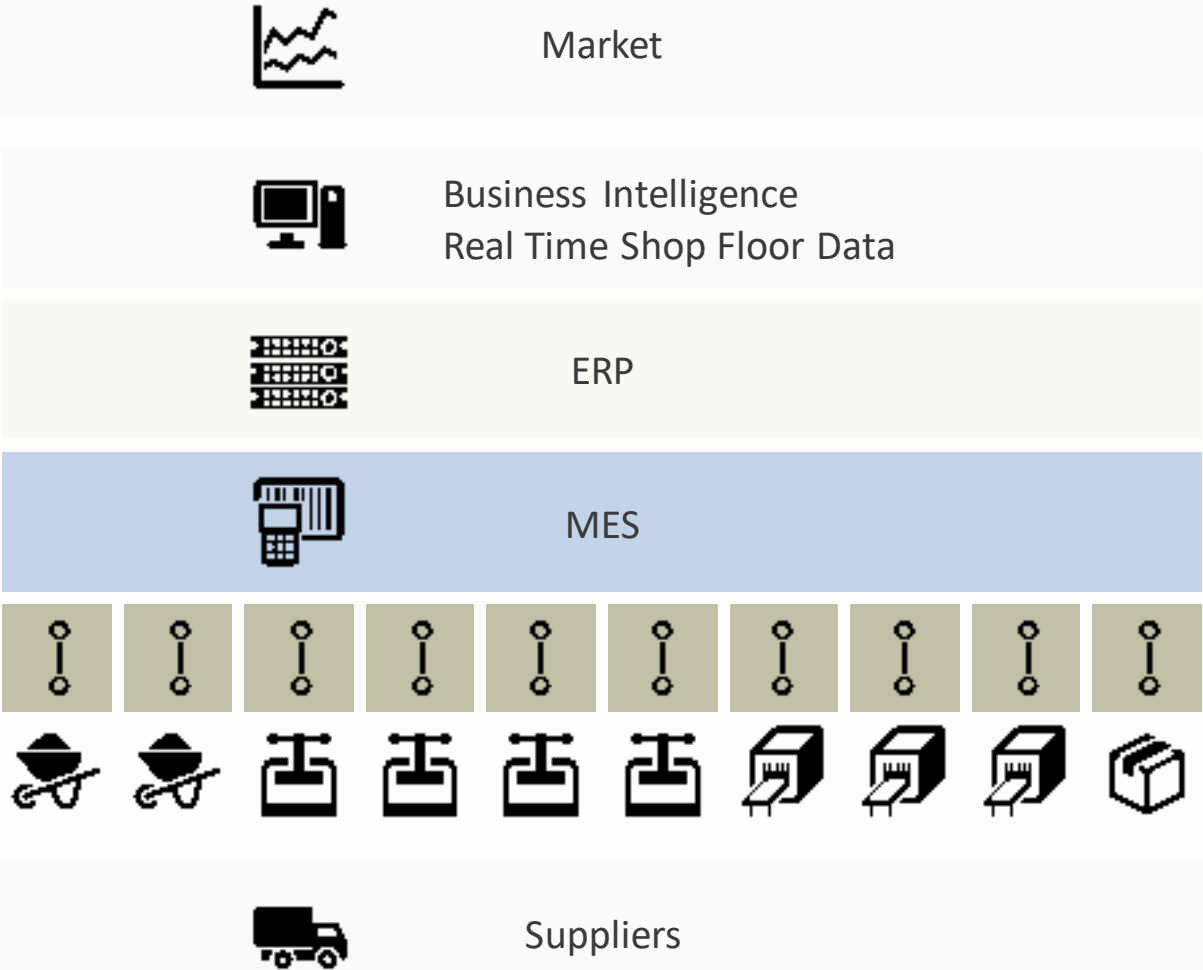
Real Time and
On Demand,
personalized
Act on Data

GKN Powder Metallurgy - A Truly Digital Driven Business

Systems | Data and Integration



Customer Request



Customer Delivery

KPIs delivered in real time to all employees across the organization

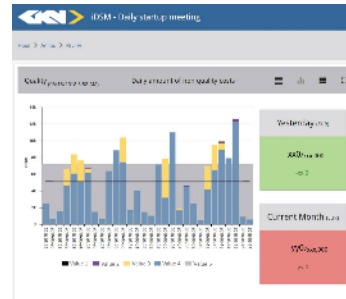
GKN PM has created a digital culture that drives value and delivers KPIs in a highly visual and easily understandable format

Visual Shop Floor



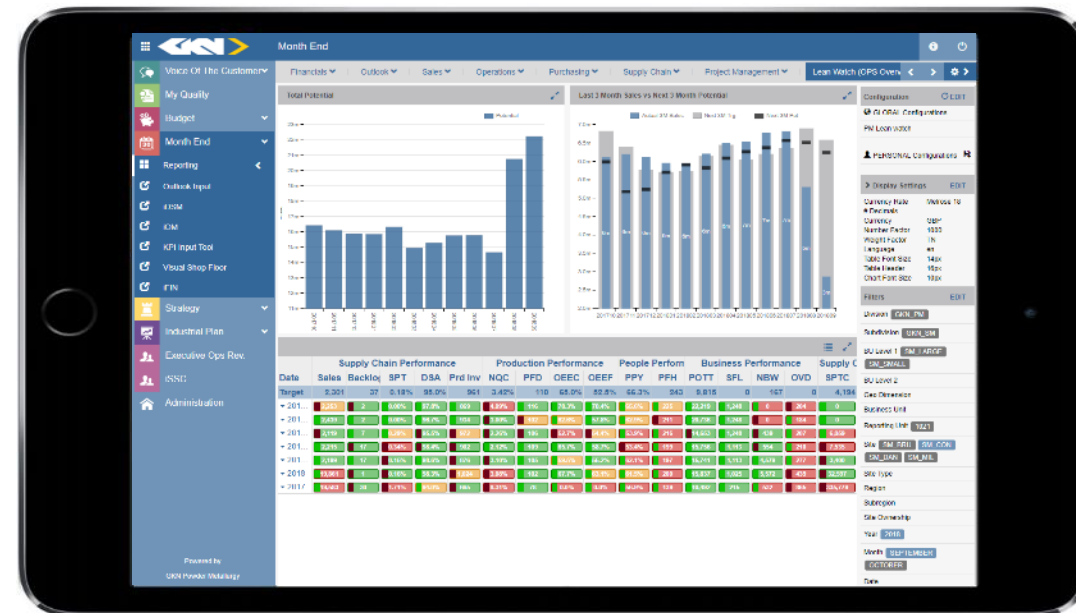
- Global insight on shop floor performance
 - Tracks all 25 plants globally
- Standardised widgets and dashboards
- Warning system for tracking critical information e.g. customer orders and machine downtimes
- Real time data and dynamic dashboards

Daily Start-up Meeting



- Digital daily start-up globally executed in every plant
- Standard agenda and set of KPIs automatically provided by the system with no manual input or change
- Execution tracked globally

Example view of KPIs delivered in real time



1

Supply Chain Performance

2

Production Performance

3

People Performance

4

Business Performance

5

Supply Chain Performance

Highlights



Real Time Data



Standardised Execution Process



Standardised approach agenda and KPIs



Global Application



Digital Manufacturing

A New Approach
Ready to change the world or to fail?

Current influences on production work – Digital Solution approaches

Demographic trends

Diversification of competence and performance
Individual support and training according to performance

Social change

Skills shortages and changes in the worklife balance
Efficient and individualized task scheduling
Flexibilisation of working time.

Customization

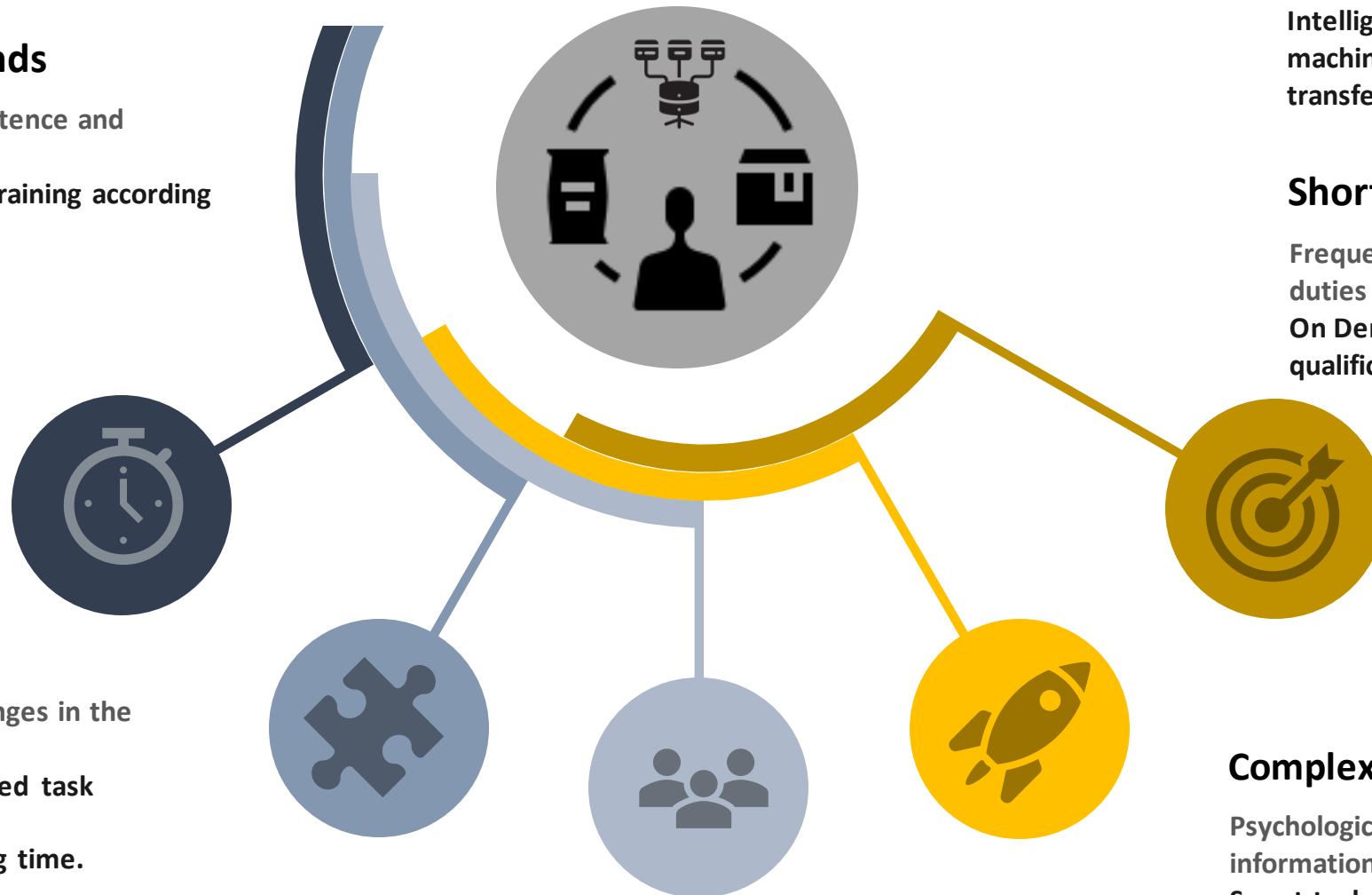
Low repetitiveness
Intelligent networking of product, machine, man and focused information transfer.

Shorter Product Life Cycle

Frequent changes of responsibilities and duties
On Demand employee oriented qualification

Complex Work Environment

Psychological stress, excessive demands, information overload
Smart technologies for decision support



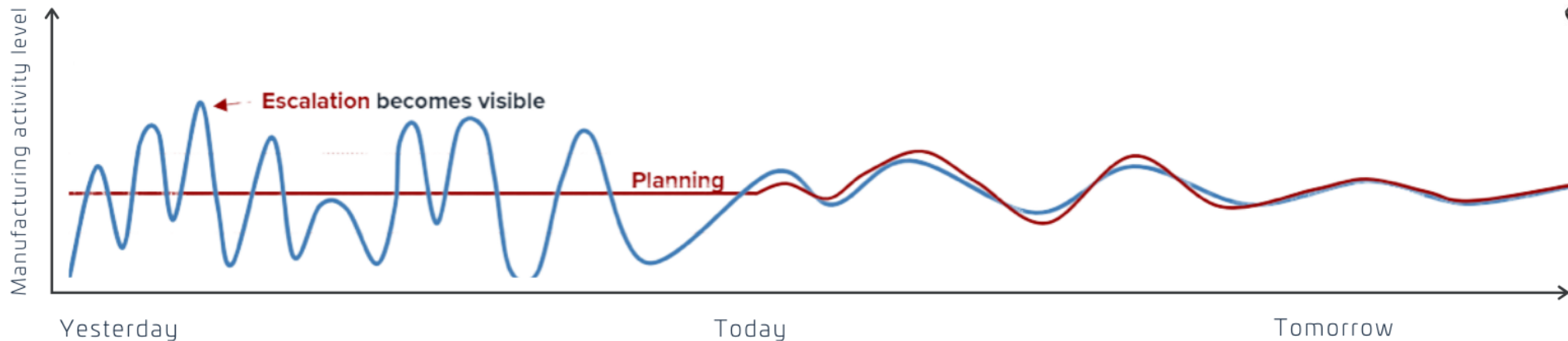
Our Journey From Lean Manufacturing to Digital Manufacturing

LEAN MANUFACTURING: STATIC RESOURCE PLANNING

➤ Machine man ratio, historic performance, experience

DIGITAL MANUFACTURING: AGILE PRODUCTION SYSTEM

➤ Dynamic planning, adaptive processes, predictive operations



REQUIREMENTS ARE CHANGING

- More dynamic market environment
- Changing customer expectations
- Data-driven ways of working
- Limited manpower and resources

WHY WE TRANSFORMED

- Dynamic shop floor management for faster processes
- Connectivity and data analytics provide process visibility
- Remote learning for planning and prediction

VALUE TO THE CUSTOMER

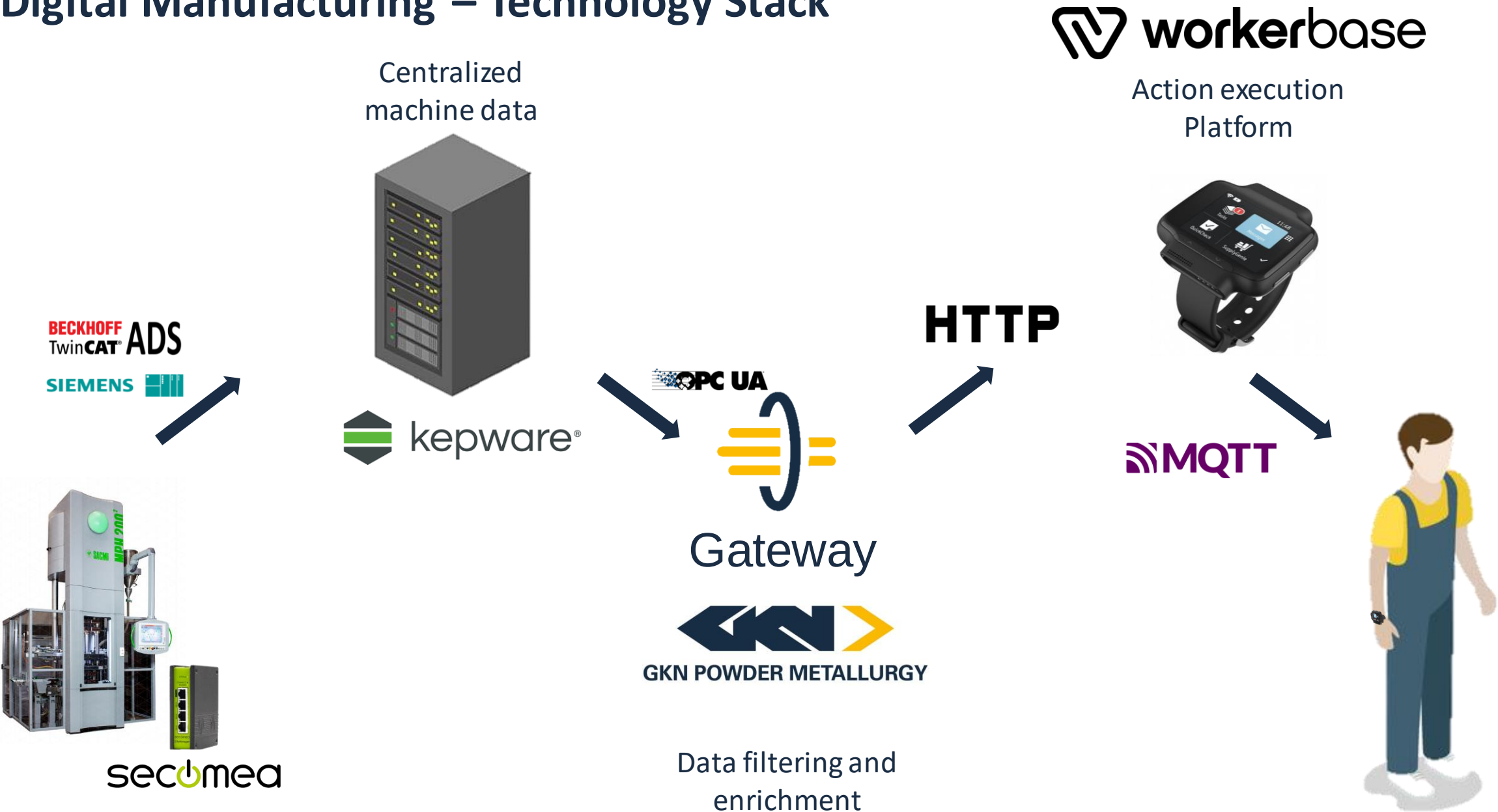
- Supply of real-time information
- Better data accuracy and transparency
- Reliable information-sharing
- Proactive supply chain



 GKN SINTER METALS



Digital Manufacturing – Technology Stack

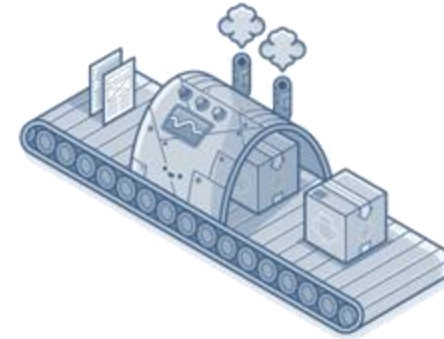


Digital Manufacturing – System KPIs



500 Users
250 daily

4 Mio Tasks
4k daily



656 Machines



2 Mio Defined
Interrupts



5 Plants

Summary and Feedback

Solid System Performance

Chaotic Feeling

Value Perception

Lack of Communication

Operator Involvement

Concept Works

Increased Stress

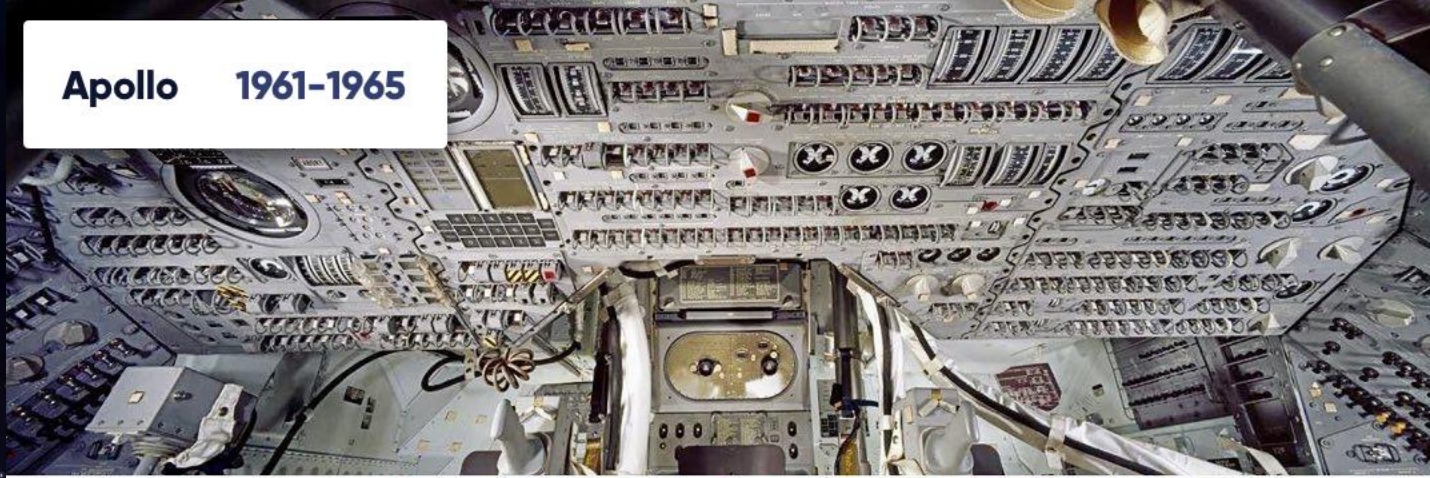
Dynamic Interaction

Speed





Apollo 1961-1965



Shuttle 1981-2011



Dragon2 2019 →









GRAND PRIX 2022

ETIHAD

ETIHAD

RACEFANS.NET

© Red Bull

How does digital Transformation work?



Change Management

Time →

WHY
Transform



WHAT
to Transform



HOW
to Transform



Change Management

Time →

WHY
Transform



WHAT
to Transform



HOW
to Transform





A person is running away from the viewer on a paved road that leads towards a bright sunset. The sun is low on the horizon, creating a strong silhouette of the runner. The sky is filled with wispy clouds, and the landscape on either side of the road is dark and rocky.

IT'S NOT A SPRINT
IT'S A MARATHON
BE PATIENT AND GRIND HARD

EVERY SINGLE DAY

THANK YOU

